THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following conclusions is NOT supported by the diffusion experiments? 1

(a) particles of matter are stationary **(b)** particles have gaps
(c) particles are in motion **(d)** temperature affects motion

2. Assertion (A): Wet clothes dry slowly on rainy days. Reason (R): The humidity of the air is high. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The ratio of the volume of water vapour to water at 100 degrees Celsius is about 1600. This large value is because: 1

(a) vapour is a liquid **(b)** particles disappear
(c) water is heavy **(d)** gas particles are far apart

4. Water in a closed vessel is heated; the pressure inside increases because: 1

(a) vapour particles collide more with the walls
(b) air is lost
(c) water becomes heavier
(d) vessel becomes smaller

5. Assertion (A): Evaporation causes cooling. Reason (R): Particles with lower energy leave the surface. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A gas cylinder gets cold when the gas is released quickly. This is because: 1

(a) pressure rises
(b) gas is cold inside
(c) liquid or compressed gas expands and absorbs heat
(d) metal is cold

7. A student heats a mixture of water and ice. A thermometer shows 0 degrees Celsius until all ice melts. Then the temperature rises. This shows: 1

(a) ice does not absorb heat **(b)** heat is first used for changing state
(c) thermometer is faulty **(d)** the water was cold

8. Assertion (A): Water at 100 degrees Celsius can exist as both liquid and vapour. Reason (R): At the boiling point both phases coexist while heat is supplied. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A sealed rigid container of gas is heated. The pressure of the gas inside: 1

(a) increases **(b)** stays the same **(c)** becomes zero **(d)** decreases

10. A drop of ink is added to equal volumes of water at 5, 25, 45 and 65 degrees Celsius. The ink spreads fastest in the water at: 1

(a) 45 degrees Celsius **(b)** 65 degrees Celsius
(c) 5 degrees Celsius **(d)** 25 degrees Celsius

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do the particles of a solid vibrate faster when it is heated? 2

12. A scientist places a sealed syringe containing air in hot water. What happens to the plunger and why? 2

13. Why is a collapsed plastic bottle left in a car in the cold seen to expand again when warm? 2

14. Why does a puddle on a cold, humid day last longer than on a hot, dry day? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A closed jar with a small amount of water at 25 degrees Celsius is kept in the sun. Explain the changes in the vapour pressure. 3

16. Why does a rubber balloon filled with air expand when taken into a warm room from a refrigerator? 3

17. Explain why wet hair dries faster when you use a hair dryer. 3

OR

Why does the sublimation of dry ice produce a white fog? 3

18. Explain why the rate of diffusion in solids is extremely slow. 3

OR

The freezer of a refrigerator forms frost. Explain it with particles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three liquids P, Q and R are placed in open dishes in identical conditions. P evaporates in 1 hour, Q in 3 hours, R in 6 hours. Rank the strength of the forces between particles and justify. Suggest which could be spirit, water and oil. 5

OR

A closed container with a movable piston holds gas. Predict and explain the changes on 5
(i) pushing the piston in
(ii) heating at constant pressure
(iii) cooling at constant volume.

20. Explain the particle model of the three states and how the model explains melting, boiling, condensation, freezing and sublimation. 5

OR

Discuss any five everyday phenomena that are explained by evaporation, condensation or sublimation and the particle model. 5

END OF QUESTION PAPER — Total: 40 Marks